MTH631 - Real Analysis II (Quiz # 3) Quiz

Question # 1 of 10 (Start time: 10:19:53 AM, 23 August 2022)

$$\text{In } \mathbb{R}^2, f(x,y) = \begin{cases} \dfrac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2<1 \\ 1, & x^2+2y^2=1 \end{cases}$$

$$\text{is ---------- on } x^2+2y^2=1\,.$$

Select the correct option

- [x] continuous

A

- [ ] discontinuous

Click to Save

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 10:19 AM, 23 Augus

Question # 2 of 10 (Start time: 10:23:12 AM, 23 August 2022) Total M

Suppose in $\mathbb{R}^2$, f, f_x, f_y and f_{xy} exist on neighborhood N of (x_0, y_0). Then $f_{yx}(x_0, y_0)$ exists, and $f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0)$ because if f_{xy} is - - - - .

Select the correct option

Reload Math Equa

- continuous
- bounded
- differntiable
- partially differentiable

B

MTH631 - Real Analysis II (Quiz # 3) Quiz

Question # 3 of 10 (Start time: 10:24:12 AM, 23 August 2022)

Let $f(x,y) = \frac{xy}{x^2+y^2}$ then the limit of f along the line $y = -x$ as (x,y) approach $(0,0)$ is _____.

Select the correct option

- ○ 0
- ○ undefined
- ○ $1/2$
- ◉ $-1/2$

Click to Sav

MTH631 - Real Analysis II (Quiz # 3)

Question # 4 of 10 (Start time: 10:25:05 AM, 23 August 2

A compact set in $\mathbb{R}^n$ is - - - - - - - - - - .

Select the correct option

○	open and bounded
○	open and unbounded
◉	closed and bounded
	C
○	closed and unbounded

MTH631 - Real Analysis II (Quiz # 3)

Question # 5 of 10 (Start time: 10:25:38 AM, 23 August 2022)

If $\phi \neq S \subseteq \mathbb{R}^n$, then the set S is bounded if - - - - - - - - .

Select the correct option

○	$\inf\{\|X-Y\| : X, Y \in S\} < \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} = \infty$
○	$\sup\{\|X-Y\| : X, Y \in S\} = \infty$
◉	$\sup\{\|X-Y\| : X, Y \in S\} < \infty$ D

Question # 7 of 10 (Start time: 10:27:31 AM, 23 August 2022)

If $\lim_{X \to X_0} f(X)$ exists, then it is _____.

Select the correct option

○	none of these
●	unique
	B
○	1
○	0

MC210401156: AZRA RAFIQUE

Time Left 70 sec(s)

MTH631 - Real Analysis II (Quiz # 3)

Quiz Start Time: 10:19 AM, 23 August 2022

Question # 8 of 10 (Start time: 10:28:04 AM, 23 August 2022)

Total Marks: 1

Intervals $(0,1)$ and $(1,2)$ are example of disconnected sets in $\mathbb{R}$ because - - - - - - - .

Select the correct option

Reload Math Equations

- $(0,1) \cap (1,2) = \phi$
- $\{\text{closure of } (0,1)\} \cap (1,2) = \phi$ and $\{\text{closure of } (1,2)\} \cap (0,1) = \phi$

B

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 10:19 AM, 23 Au

Question # 10 of 10 (Start time: 10:29:15 AM, 23 August 2022) To

In $\mathbb{R}^n$, which of the following is true about $\phi = \{\}$ and $A = \{(a_1, a_2, \ldots, a_n), a_i \in \mathbb{R}, 1 \leqslant i \leqslant n, i \in \mathbb{N}\}$?

Select the correct option

Reload Math E

- Both are Disconnected
- ϕ is Disconnected and A is Connected
- ϕ is Connected and A is Disconnected
- Both are Connected

D

Click to Save Answer & Move to Next Q

MTH631 - Real Analysis II (Quiz # 3) Quiz St

Question # 2 of 10 (Start time: 10:35:57 AM, 23 August 2022)

$$\text{In } \mathbb{R}^2, f(x,y) = \begin{cases} \dfrac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2<1 \\ 1, & x^2+2y^2=1 \end{cases}$$

is - - - - - - - - - - on $x^2+2y^2=1$.

Select the correct option

- (A) continuous
- discontinuous

Click to Save

3 August 2022)

$$\lim_{(x,y)\to(a,b)} \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}} = 1, \text{ if} \text{ - - - - - - .}$$

$(a^2+2b^2) > 1$

B $(a^2+2b^2) = 1$

$(a^2+2b^2) < 1$

$(a^2+2b^2) \geq 1$

(Quiz # 3) Quiz Start Tim

ime: 10:40:01 AM, 23 August 2022)

In $\mathbb{R}^n$, monotonicity, limits inferior and superior of sequences, and divergence to $\pm\infty$ are undefined for $n > 1$ because $\mathbb{R}^n$ is - - - - - - - - .

complete

not compact for $n > 1$

C not ordered for $n > 1$

a Field

Click to Save Answer

Q

M, 23 August 2022)

In $\mathbb{R}^2$, $\lim_{(x,y)\to(2,2)} \frac{\sin(x-y)}{\sqrt{x-y}} = ----$.

∞

1

$\sqrt{2}$

D 0

Click to

MTH631 - Real Analysis II (Quiz # 3)

Question # 7 of 10 (Start time: 10:41:20 AM, 23 August 2022)

If f is continuous on a compact set S in $\mathbb{R}^n$, then f is _____ on S.

Select the correct option

○	All of these
◉	Bounded B
○	Differentiable
○	Superemum

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 1

Question # 8 of 10 (Start time: 10:42:06 AM, 23 August 2022)

The domain of the real valued function $f : \mathbb{R}^2 \to \mathbb{R}$ defined and given by; $f(X) = \dfrac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}$ is - - - - - of the region by the ellipse $x^2 + 2y^2 = 1$.

Select the correct option

- exterior
- interior
- interior and boundary
- exterior and boundary

B

Click to Save Answer & M

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 10:35 AM, 23 Aug

Question # 10 of 10 (Start time: 10:44:45 AM, 23 August 2022) Tota

If $f(s) = \sqrt{s}$ and $g(x, y) = 1 - x^2 - y^2$ then domain of f and g are _____.

Select the correct option

Reload Math Eq

- $D_f = [0, \infty), D_g = \mathbb{R}^2$
- $D_g = (-\infty, \infty), D_f = \mathbb{R}^2$
- $D_f = [0, \infty], D_g = \mathbb{R}^2$
- $D_f = [0, \infty], D_g = \mathbb{R}$

C

Click to Save Answer & Move to Next Qu

MTH631 - Real Analysis II (Quiz # 3) Quiz

Question # 6 of 10 (Start time: 10:26:23 AM, 23 August 2022)

Let the function $f(s) = \begin{cases} \frac{\sin s}{s}, s \neq 0 \\ 1, s = 0 \end{cases}$ is continuous over domain _____.

Select the correct option

○	(c) $(0, \infty)$
○	(a) $\mathbb{R}$
○	Both a & b
○	(b) $(-\infty, \infty)$

D

Click to Sav

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 11:34 AM, 23 Augu

Question # 1 of 10 (Start time: 11:34:35 AM, 23 August 2022) Total

Let f be defined and continuous on a region S in $\mathbb{R}^n$. Suppose that X_1 and X_2 are in S and $f(X_1) < Y < f(X_2)$.

Then $f(X) = Y$ - - - - - - - X in S.

Select the correct option

Reload Math Equ

- for some
- for all

A

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 11:34 AM, 23 A

Question # 2 of 10 (Start time: 11:35:45 AM, 23 August 2022) To

In $\mathbb{R}^n$, if $S_1, S_2, S_3 \ldots$ are non - empty closed subsets such that $S_1 \supset S_2 \supset \cdots \supset S_r \supset \cdots$ and $\sup\{|X_r - Y_r| : X_r, Y_r \in S_r, r \geqslant 1\} \to 0$ as $r \to \infty$,then the order of set $\bigcap_{r=1}^{\infty} S_r = - - -$.

Select the correct option

Reload Math E

- infinite
- finite but more than one
- one
- zero

C

Click to Save Answer & Move to Next Q

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Tin

Question # 3 of 10 (Start time: 11:36:11 AM, 23 August 2022)

If f is continuous on a compact set S in $\mathbb{R}^n$, then f – – – – on S.

Select the correct option

◉	is also uniformly continuous A
○	is also defined on all the limit points of "S"
○	attains all its bounds
○	All above are equally valid

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time:

Question # 4 of 10 (Start time: 11:36:59 AM, 23 August 2022)

The domain of the real valued function $f : \mathbb{R}^2 \to \mathbb{R}$ defined and given by;

$$f(X) = \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}$$ is - - - - - of the region by the ellipse $x^2 + 2y^2 = 1$.

Select the correct option

- interior and boundary
- exterior and boundary
- exterior
- interior

D

Click to Save Answer & M

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time

Question # 5 of 10 (Start time: 11:37:47 AM, 23 August 2022)

In $\mathbb{R}^n$, monotonicity, limits inferior and superior of sequences, and divergence to $\pm\infty$ are undefined for $n > 1$ because $\mathbb{R}^n$ is - - - - - - - - .

Select the correct option

○	not compact for $n > 1$
●	not ordered for $n > 1$
○	complete
○	a Field

B

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 11:

Question # 7 of 10 (Start time: 11:38:59 AM, 23 August 2022)

In $\mathbb{R}^2$, the set $\{(x,y) : (x^2+y^2 \leqslant a) \vee (x^2+y^2 \geqslant b), a < b\}$ is a region.

Select the correct option

Re

- (●) False
- ( ) True

A

Click to Save Answer & Ma

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 11:34 AM, 23 August 2022

Question # 8 of 10 (Start time: 11:40:16 AM, 23 August 2022) Total Marks: 1

In $\mathbb{R}^n$, which of the following is true about $\phi = \{\}$ and $A = \{(a_1, a_2, \ldots, a_n), a_i \in \mathbb{R}, 1 \leqslant i \leqslant n, i \in \mathbb{N}\}$?

Select the correct option

Reload Math Equations

- ϕ is Connected and A is Disconnected
- Both are Connected
- Both are Disconnected
- ϕ is Disconnected and A is Connected

B

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 11:

Question # 9 of 10 (Start time: 11:42:28 AM, 23 August 2022)

If a function f is continuous on a Compact set S in $\mathbb{R}^n$, then f is - - - - - - - - on S.

Select the correct option

Re

- (●) bounded — A
- ( ) bounded above
- ( ) bounded below
- ( ) unbounded

Click to Save Answer & Ma

MTH631 - Real Analysis II (Quiz # 3)

Question # 10 of 10 (Start time: 11:43:04 AM, 23 August 2022)

If $f(x, y, z) = \cos\left(\frac{1}{x^2+2y^2+z^2}\right)$ then $\lim_{|X|\to\infty} f(X) =$ _____

Select the correct option

○	0
◉	infinite
○	1
○	-1

C

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 11:34 AM, 23 August 2022

Question # 6 of 10 (Start time: 11:38:37 AM, 23 August 2022) Total Marks: 1

The limit of the function $f(x) = \frac{xy}{x^2+y^2}$ by letting (x, y)approach $(0, 0)$along the line $y = x$ is _____.

Select the correct option

Reload Math Equations

- Finite
- $\frac{1}{2}$
- Undefined
- $-\frac{1}{2}$

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 11:50 AM, 23 August 2022

Question # 6 of 10 (Start time: 11:52:28 AM, 23 August 2022) Total Marks: 1

If $\mathbb{R}^n$ is connected, such that $\mathbb{R}^n = A \cup B$ with $\bar{A} \cap B = A \cap \bar{B} = \phi$, then - - - - - .

Select the correct option

Reload Math Equations

○	$\bar{A} \subset A, \bar{B} \subset B$
○	All above can be concluded
○	A and B are both open and closed
○	Either $A = \mathbb{R}^n, B = \phi$ or $B = \mathbb{R}^n, A = \phi$

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3)

Question # 9 of 10 (Start time: 11:53:38 AM, 23 August 2022)

An open disc: $\{(x, y) : x^2 + y^2 < 1\}$ in $\mathbb{R}^2$ is - - - - - - .

Select the correct option

○	connected polygonally
○	none of these.
○	disconnected polygonally
○	diconnected

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 12:06 PM, 23 August 2022

Question # 1 of 10 (Start time: 12:06:00 PM, 23 August 2022) Total Marks: 1

In $\mathbb{R}^n$, $f(X) = \frac{1}{|X - X_0|}$, $and \lim_{X \to X_0} f(X) = \infty$, then $f(X) > M > 0 \Rightarrow 0 < |X - X_0| < \delta = ----$.

Select the correct option

Reload Math Equations

○	$\frac{1}{\sqrt{M}}$
○	$\sqrt{M}$
◉	$\frac{1}{M}$ C
○	M

Click to Save Answer & Move to Next Question

23 August 2022)

In $\mathbb{R}^n$, $\lim_{X \to X_0} f(X) = -\infty$, if

$\lim_{X \to X_0} (-f)(X) = \infty$

$\lim_{-X \to X_0} f(X) = \infty$

$\lim_{X \to -X_0} f(X) = \infty$

$\lim_{X \to X_0} f(-X) = \infty$

: # 3)

Quiz Start T

12:07:37 PM, 23 August 2022)

In $\mathbb{R}^2$, $f(X) = |X|^2$ and $\Phi = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, then $\frac{\partial f(X)}{\partial \Phi} =$

A $\sqrt{2}(x_1 + x_2)$

$x_1 + x_2$

$\frac{x_1 + x_2}{\sqrt{2}}$

$\frac{x_1 + x_2}{2}$

Click to Save Answ

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 12:06 PM, 23 August 2022

Question # 4 of 10 (Start time: 12:08:18 PM, 23 August 2022) Total Marks: 1

If $\phi \neq S \subseteq \mathbb{R}^n$, then the set S is bounded if - - - - - - - .

Select the correct option

Reload Math Equations

◉	$\sup\{\|X-Y\| : X, Y \in S\} < \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} < \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} = \infty$
○	$\sup\{\|X-Y\| : X, Y \in S\} = \infty$

A

Saving...

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 12:06 PM, 23 August 2022

Question # 5 of 10 (Start time: 12:08:56 PM, 23 August 2022) Total Marks: 1

$$\lim_{(x,y)\to(a,b)} \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}} = 1, \text{ if - - - - - - .}$$

Select the correct option

Reload Math Equations

- A $(a^2+2b^2)=1$
- $(a^2+2b^2)\geq 1$
- $(a^2+2b^2)<1$
- $(a^2+2b^2)>1$

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 12:06 PM, 23 Augus

Question # 7 of 10 (Start time: 12:10:16 PM, 23 August 2022) Total M

Suppose in $\mathbb{R}^2$, f, f_x, f_y and f_{xy} exist on neighborhood N of (x_0, y_0). Then $f_{yx}(x_0, y_0)$ exists, and $f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0)$ because if f_{xy} is - - - - .

Select the correct option

Reload Math Equa

- differntiable
- partially differentiable
- bounded
- continuous

D

Click to Save Answer & Move to Next Quest

MTH631 - Real Analysis II (Quiz # 3) Quiz Start T

Question # 8 of 10 (Start time: 12:13:48 PM, 23 August 2022)

$$\text{In } \mathbb{R}^2, f(x,y) = \begin{cases} \dfrac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2<1 \\ 1, & x^2+2y^2=1 \end{cases}$$

is ---------- on $x^2+2y^2=1$.

Select the correct option

- discontinuous
- continuous

B

Click to Save Answ

631 - Real Analysis II (Quiz # 3) Quiz Start Time: 12:06 PM,

tion # 9 of 10 (Start time: 12:14:32 PM, 23 August 2022)

Let f be defined on an interval I in R^2. Suppose that x_1 and x_2 are in I and $f(x_1) < y < f(x_2)$. Then $f(x) = y$ for some x in I.

t the correct option

Reload M

False

True

B

Click to Save Answer & Move to N

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 12:06 PM

Question # 10 of 10 (Start time: 12:15:03 PM, 23 August 2022)

In $\mathbb{R}^n$, the function $f(X)$ is continuous at X_0, if $\lim_{X \to X_0} f(X) = f(X_0)$, then - - - - - - .

Select the correct option

Reload

- X_0 is in the domian D_f and limit point of D_f
- X_0 is in the domian D_f but not a limit point of D_f
- X_0 is not in the domian D_f but a limit point of D_f
- neither X_0 is in the domian D_f nor the limit point of D_f

Click to Save Answer & Move to

Question # 5 of 10 (Start time: 09:58:32 AM, 24 August 2022)

Quiz Start Time: 09:58 AM, 24 August 20

Total Marks

Analogous to the derivative of a function of one variable in $\mathbb{R}^2$, the directional derivative of a function f at X_0 in $\mathbb{R}^n$ has - - - - - - value(s).

Select the correct option

Reload Math Equations

- integral
- unique
- infinite many
- multiple

C

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 10:54 AM, 24 August 2022

Question # 2 of 10 (Start time: 10:54:51 AM, 24 August 2022) Total Marks: 1

If $g(x,y)=\sqrt{1-x^2-2y^2}$, $f(t)=\begin{cases}\frac{\sin t}{t}, & t\neq 0\\ 1, & t=0\end{cases}$,

then domain of $f\circ g=-----$.

Select the correct option

Reload Math Equations

○	$\mathbb{R}$
○	$\{(x,y): x^2-2y^2\leqslant 1\}$
○	$\mathbb{R}^2$
○	$\{(x,y): x^2-2y^2\geqslant 1\}$

B

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3)

Quiz Start Time: 10:54 AM, 24 August 2022

Question # 7 of 10 (Start time: 10:58:32 AM, 24 August 2022)

Total Marks: 1

Intervals $(0,1)$ and $(1,2)$ are example of disconnected sets in $\mathbb{R}$ because - - - - - - - .

Select the correct option

Reload Math Equations

- $\{\text{closure of } (0,1)\} \cap (1,2) = \phi$ and $\{\text{closure of } (1,2)\} \cap (0,1) = \phi$
- $(0,1) \cap (1,2) = \phi$

A

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 10:54 AM, 24 August 2022

Question # 10 of 10 (Start time: 11:00:37 AM, 24 August 2022) Total Marks: 1

If a function f is continuous on a Compact set S in $\mathbb{R}^n$, then f is - - - - - - - - on S.

Select the correct option

Reload Math Equations

- bounded above
- unbounded
- bounded below
- bounded

D

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 11:12:48 AM, 24 August 2022)

If $f(s) = \sqrt{s}$ and $g(x, y) = 1 - x^2 - y^2$ then domain of f and g are _____

Select the correct option

○	$D_g = (-\infty, \infty), D_f = \mathbb{R}^2$
○	$D_f = [0, \infty], D_g = \mathbb{R}$
○	$D_f = [0, \infty], D_g = \mathbb{R}^2$
○	$D_f = [0, \infty), D_g = \mathbb{R}^2$

C

MTH631:Quiz-2

Question # 3 of 10 (Start time: 09:54:16 AM, 18 February 2021)

Sequence in $\mathbb{R}^2$ defined as;

$\{X_r\} = \left\{\left(\frac{\cos r}{r}, \frac{\sin r}{r}\right)\right\}$ converges to - - - - - .

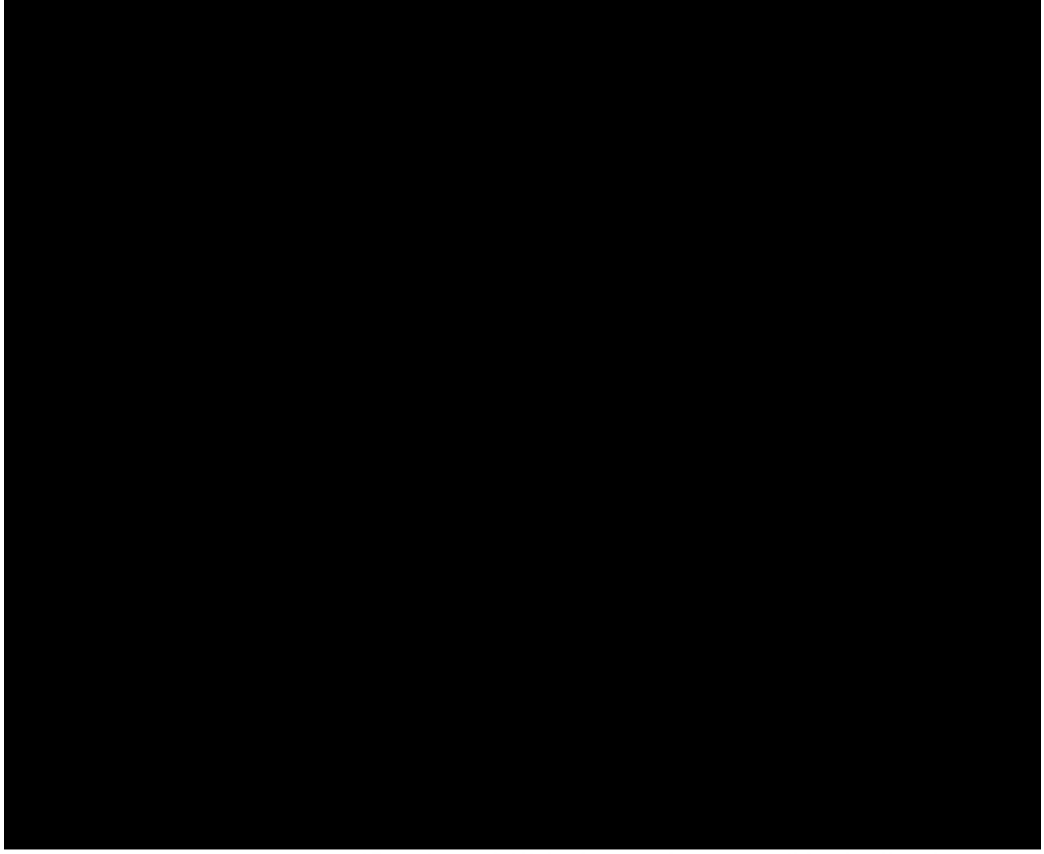
Select the correct option

- $(1, 0)$
- $(0, 1)$
- $(0, 0)$ C
- $(1, 1)$

In $\mathbb{R}^n$, if the line segment: $X = tX_2 + (1-t)X_1,\ 0 < t < 1$, joining X_1 and X_2 lies in $S_\epsilon(X_0)$, then ----.

Select the correct option

- A $|X - X_0| < \epsilon$
- $|X - X_0| < 1 - \epsilon$
- $|X - X_0| < t$
- $|X - X_0| < 1 - t$



How many *third* order partial derivatives of $g(x, y) = xy + x^2y^3$ exist in $\mathbb{R}^3$?

ect the correct option

Reload

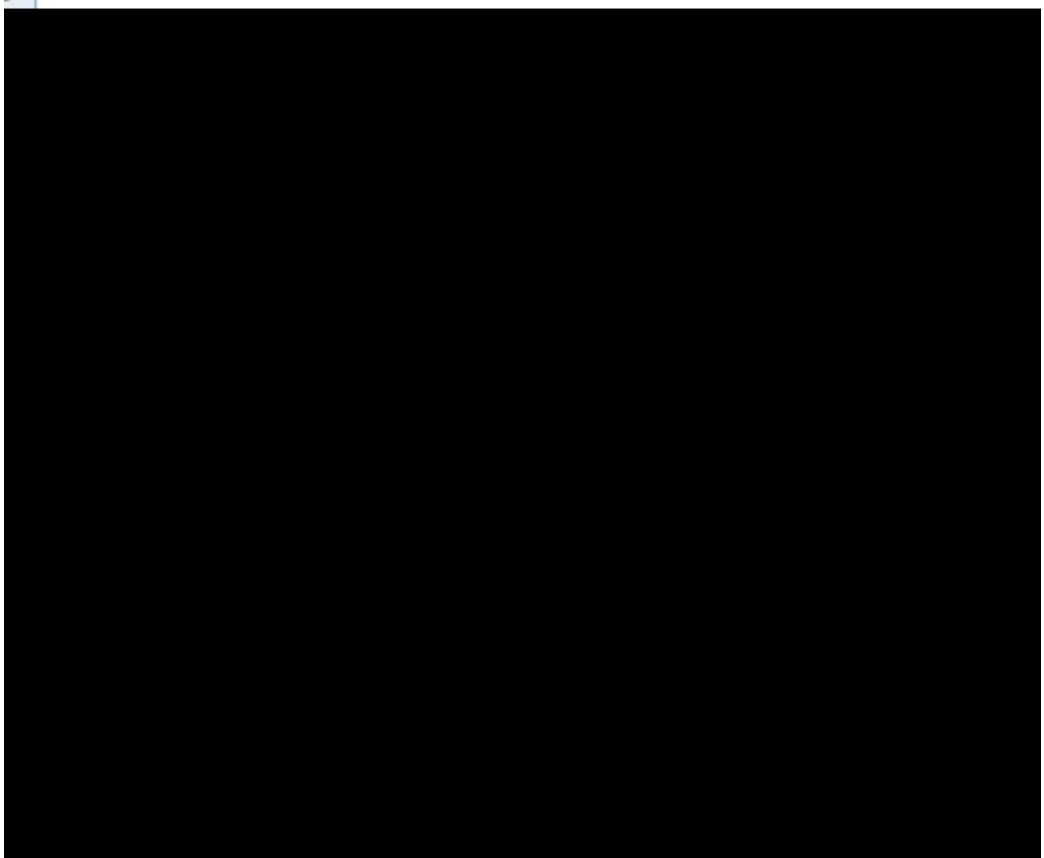
3

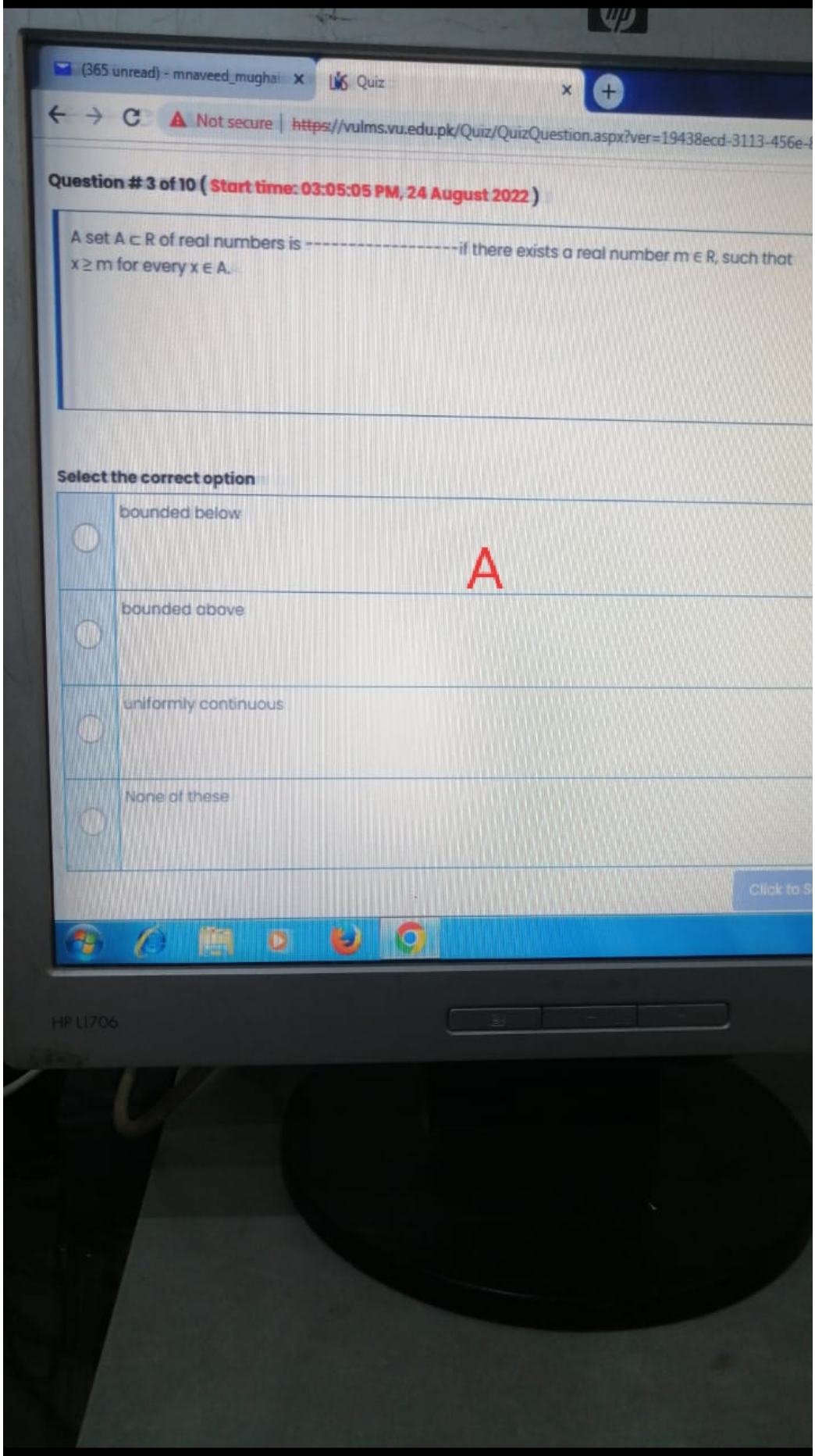
6

8

C

9




(365 unread) - mnaveed_mughal
Quiz
Not secure
https://vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=19438ecd-3113-456e-
Question # 3 of 10 (Start time: 03:05:05 PM, 24 August 2022)
A set A ⊂ R of real numbers is ---------- if there exists a real number m ∈ R, such that x ≥ m for every x ∈ A.
Select the correct option
bounded below
A
bounded above
uniformly continuous
None of these
Click to S
HP LI706

, 25 August 2022)

If $g(x,y)=\sqrt{1-x^2-2y^2}$, $f(t)=\begin{cases}\frac{\sin t}{t}, & t\neq 0\\ 1, & t=0\end{cases}$,

then domain of $f\circ g=$ ------.

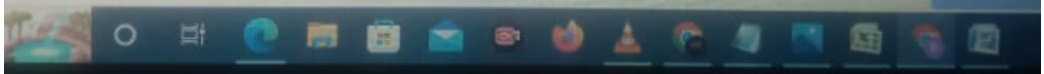
$\{(x,y):x^2-2y^2\geqslant 1\}$

$\{(x,y):x^2-2y^2\leqslant 1\}$

B

$\mathbb{R}^2$

$\mathbb{R}$



11:12 24.26 KB/s 4G
vulms.vu.edu.pk/Quiz/

MC210202213: MUHAMMAD ZEESHAN

Time Left 90 sec(s)

MTH631 - Real Analysis II (Quiz # 3)

Quiz Start Time: 11:08 AM, 24 August 2022

Question # 10 of 10 (Start time: 11:12:48 AM, 24 August 2022)

Total Marks: 1

If $f(s) = \sqrt{s}$ and $g(x, y) = 1 - x^2 - y^2$ then domain of f and g are _____.

Select the correct option

Reload Math Equations

- $D_g = (-\infty, \infty), D_f = \mathbb{R}^2$
- $D_f = [0, \infty], D_g = \mathbb{R}$
- $D_f = [0, \infty], D_g = \mathbb{R}^2$
- $D_f = [0, \infty), D_g = \mathbb{R}^2$

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3)

Quiz Start Time: 07:10 PM, 23 August 2022

Question # 2 of 10 (Start time: 07:11:16 PM, 23 August 2022)

Total Marks: 1

A function f is _____ on a subset S of its domain in $\mathbb{R}^n$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(X) - f(X')| < \varepsilon$ whenever $|X - X'| < \delta$ and $X, X' \in S$.

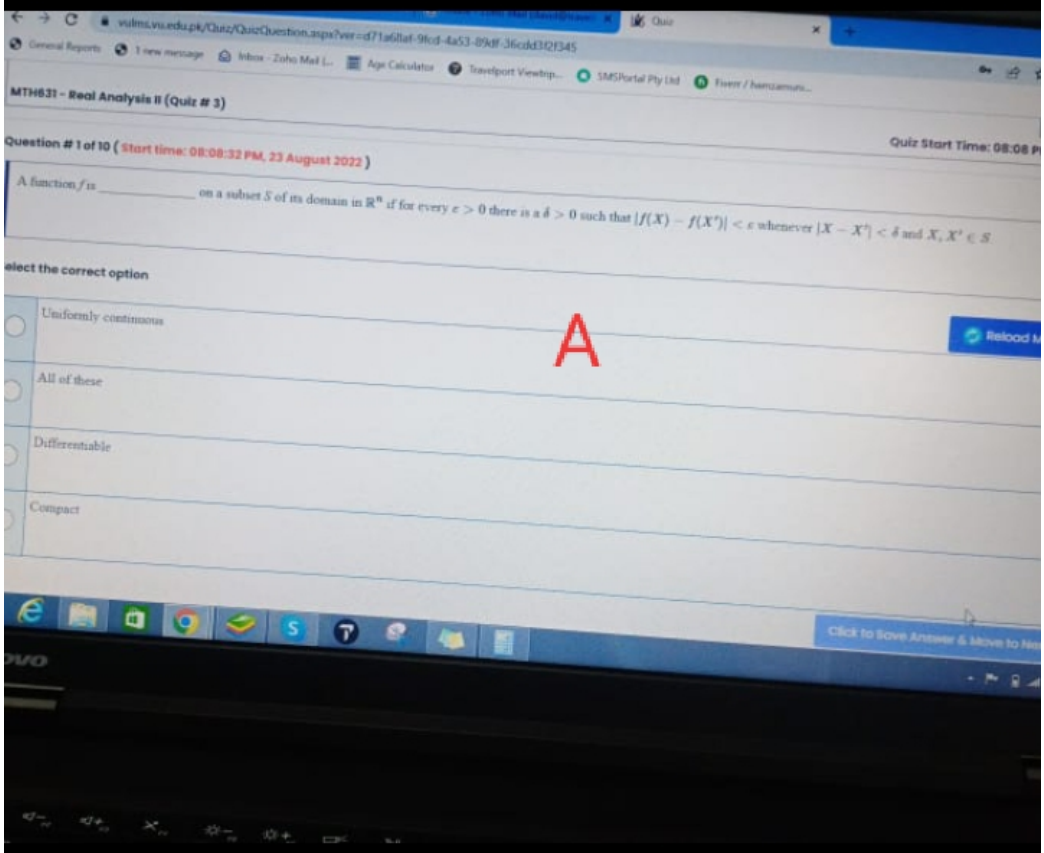
Select the correct option

Reload Math Equations

- Differentiable
- All of these
- Compact
- Uniformly continuous

D

Click to Save Answer & Move to Next Question


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Quiz
MTH631 - Real Analysis II (Quiz # 3)
Question # 1 of 10 (Start time: 08:08:32 PM, 23 August 2022)
Quiz Start Time: 08:08
A function f is _____ on a subset S of its domain in $\mathbb{R}^n$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(X) - f(X')| < \varepsilon$ whenever $|X - X'| < \delta$ and $X, X' \in S$
elect the correct option
Uniformly continuous
A
Reload
All of these
Differentiable
Compact
Click to Save Answer & Move to Ne

MC210401828: SAIRA BEGUM

MTH631 – Real Analysis II (Quiz # 3)

Question # 8 of 10 (Start time: 06:55:52 PM, 23 August 2022)

Let f be continuous on a region S in $\mathbb{R}^n$. Suppose that A and B are in S and _____. Then $f(C) = u$ for some C in S.

Select the correct option

- $f(A) \leq u \leq f(B)$
- $f(A) > u > f(B)$
- $f(A) \geq u \geq f(B)$
- $f(A) < u < f(B)$

D

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 07:24 PM, 23 August 2022

Question # 1 of 10 (Start time: 07:24:14 PM, 23 August 2022) Total Marks: 1

In $\mathbb{R}^n$, the interior S^0 of a non - empty set S is - - - - - - .

Select the correct option

Reload Math Equations

- open - connected (region)
- open - disconnected

A

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 07:24 PM, 23 August 2022

Question # 3 of 10 (Start time: 07:25:08 PM, 23 August 2022) Total Marks: 1

A set " S " is polygonally connected if, -------- pair of points in S can be connected by a polygonal path lying -------- in " S ".

Select the correct option

- some, entirely
- some, partially
- every , entirely
- every, partially

C

Click to Save Answer & Move to Next Question



MTH631:Quiz-2

Quiz Start Time: 09:59 AM, 18 February 2021

Question # 5 of 10 (Start time: 09:59:29 AM, 18 February 2021)

Total Marks: 1

A compact set in $\mathbb{R}^n$ is - - - - - - - - - - .

Select the correct option

Reload Math Equations

- open and unbounded
- open and bounded
- closed and unbounded
- closed and bounded

Ad

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 07:24 PM, 23 August 2022

Question # 5 of 10 (Start time: 07:25:50 PM, 23 August 2022) Total Marks: 1

Suppose in $\mathbb{R}^2$, f, f_x, f_y and f_{xy} exist on neighborhood N of (x_0, y_0). Then $f_{yx}(x_0, y_0)$ exists, and $f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0)$ because if f_{xy} is - - - - .

Select the correct option

Reload Math Equations

- differntiable
- continuous
- partially differentiable
- bounded

B

Click to Save Answer & Move to Next Question

MTH631 - Real Analysis II (Quiz # 3) Quiz Start Time: 10:19 AM, 23 August 2022

Question # 9 of 10 (Start time: 10:28:33 AM, 23 August 2022) Total Marks: 1

The domain of the real valued function $f : \mathbb{R}^n \to \mathbb{R}$ defined and given by;
$f(X) = \left(1 - x_1^2 - x_2^2 - \cdots - x_2^n\right)^{-1}$ is - - - - - .

Select the correct option

Reload Math Equations

- (●) $\{X \mid |X| = 1\}$
- ( ) $\{X \mid |X| \neq 1\}$

B

Click to Save Answer & Move to Next Question

10:27 4G 57

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MC210401156: AZRA RAFIQUE

Time Left 41 sec(s)

MTH631 - Real Analysis II (Quiz # 3)

Quiz Start Time: 10:19 AM, 23 August 2022

Question # 6 of 10 (Start time: 10:26:23 AM, 23 August 2022)

Total Marks: 1

Let the function $f(s) = \begin{cases} \frac{\sin s}{s}, s \neq 0 \\ 1, s = 0 \end{cases}$ is continuous over domain _____.

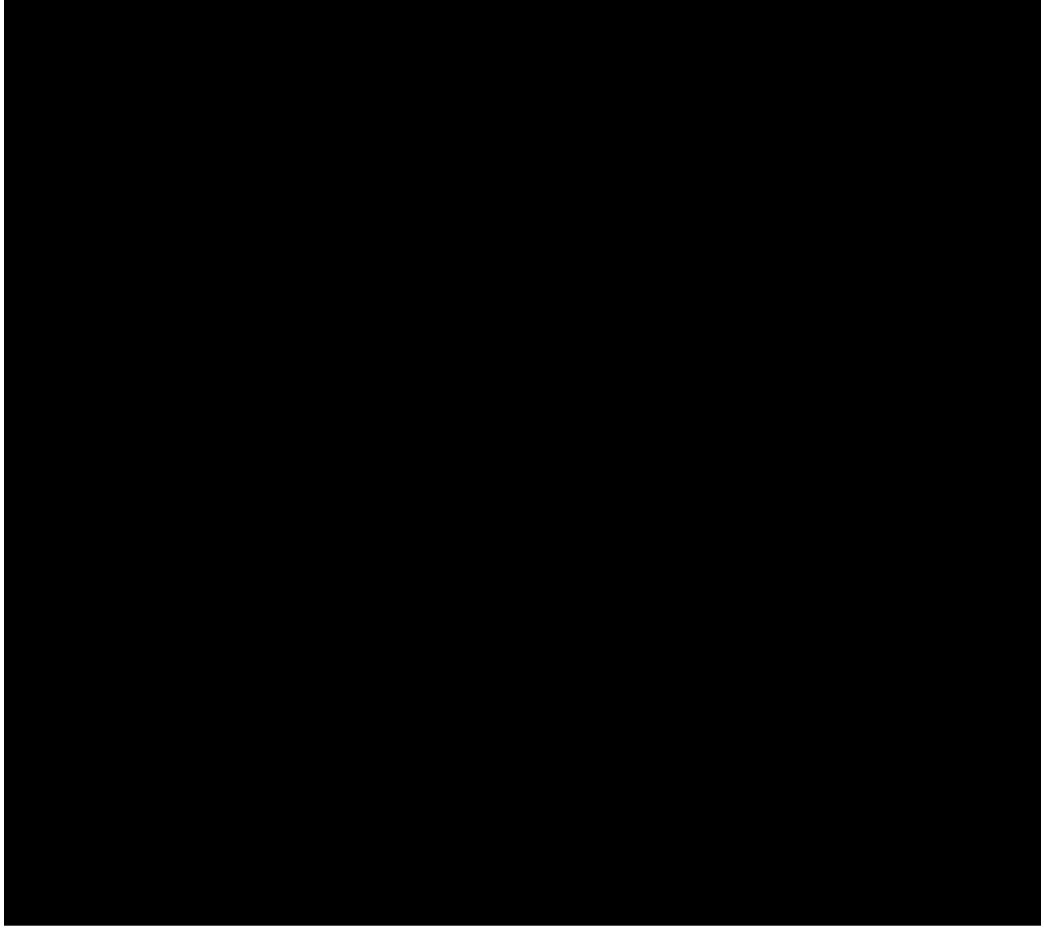
Select the correct option

Reload Math Equations

◉	(c) $(0, \infty)$
○	(a) $\mathbb{R}$
○	Both a & b
○	(b) $(-\infty, \infty)$

D

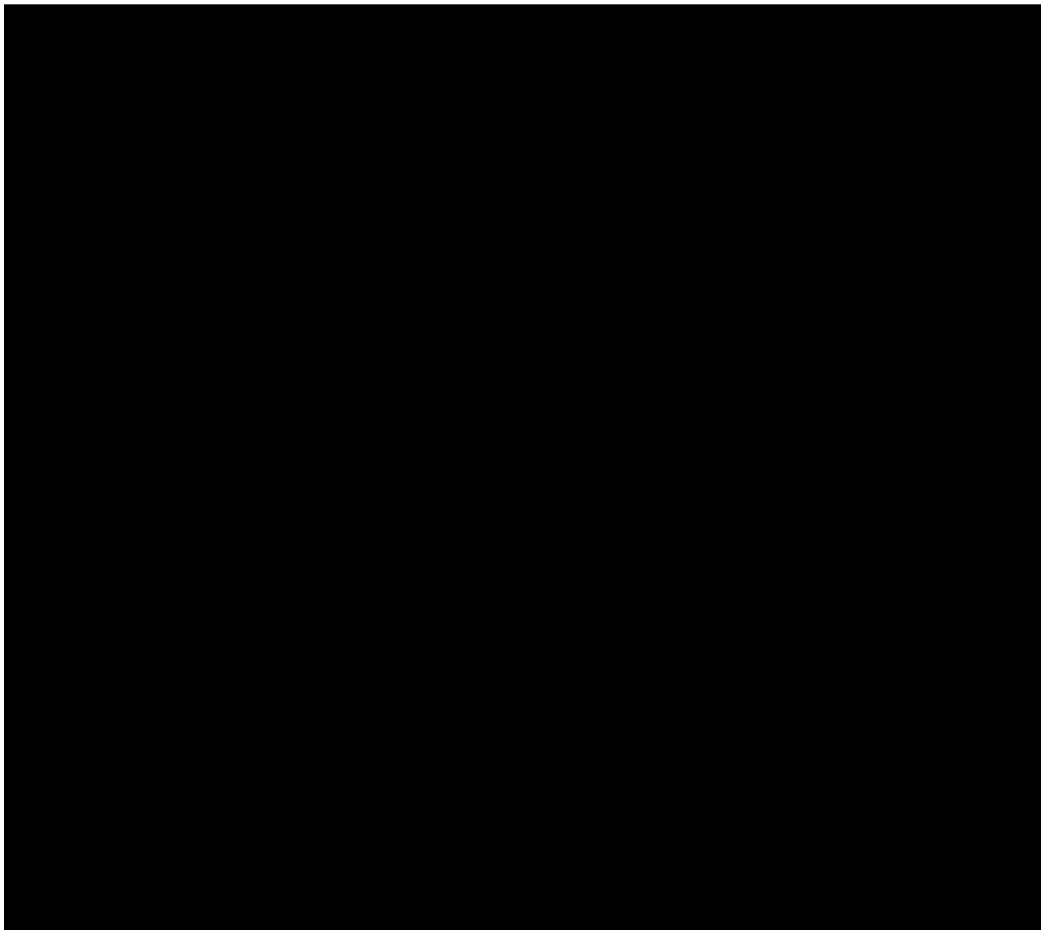
Click to Save Answer & Move to Next Question



$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = -\infty$$

$$if$$

$$\lim_{\mathbf{X}\to\mathbf{X}_0} (-f)(\mathbf{X}) = \infty.$$



Remark: Any polygonally connected set, open or not, is connected. The converse is false. A set (not open) may be connected but not polygonally connected.

Regions in $\mathbb{R}^n$: A *region* S in $\mathbb{R}^n$ is the union of an open connected set with some, all, or none of its boundary; thus, S^0 is connected, and every point of S is a limit point of S^0.

Example: Intervals are the only regions in $\mathbb{R}$. The n-ball $B_r(\mathbf{X}_0)$ is a region in $\mathbb{R}^n$, as is its closure $\overline{S}_r(\mathbf{X}_0)$. The set $S = \{(x, y) : x^2 + y^2 \leq 1 \quad \text{or} \quad x^2 + y^2 \geq 4\}$ is not a region in $\mathbb{R}^2$, since it is not connected.

The set S_1 obtained by adding the line segment

2.9 Sequences in $\mathbb{R}^n$

A sequence $\{\mathbf{X}_r\}$ of points in $\mathbb{R}^n$ converges to a limit $\overline{\mathbf{X}}$ if and only if for every $\varepsilon > 0$ there is an integer K such that

$$|\mathbf{X}_r - \overline{\mathbf{X}}| < \varepsilon \quad \text{if} \quad r \geq K.$$

The $\mathbb{R}^n$ definitions of divergence, boundedness, subsequence, and sums, differences, and constant multiples of sequences are analogous to those we discussed in Analysis I.

Since $\mathbb{R}^n$ is not ordered for $n > 1$, monotonicity, limits inferior and superior of sequences in $\mathbb{R}^n$, and divergence to $\pm\infty$ are undefined for $n > 1$.

Example: The function

$$h(x,y) = \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}$$

is defined only on the interior of the region bounded by the ellipse

$$x^2 + 2y^2 = 1.$$

It is not defined at any point of the ellipse itself or on any deleted neighborhood of

Example: The function

$$f(x,y) = \frac{xy}{x^2+y^2}$$

is defined everywhere in $\mathbb{R}^2$ except at $(0,0)$. Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

If we try to answer this question by letting (x,y) approach $(0,0)$ along the $y = x$, we see the functional values

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2}$$

and conclude that the limit is $1/2$.

However, if we let (x, y) approach $(0, 0)$ along the line $y = -x$, we see the functional values

$$f(x, -x) = -\frac{x^2}{2x^2} = -\frac{1}{2}$$

and conclude that the limit equals $-1/2$.

$$\begin{aligned} \lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) &= -\infty \\ &if \\ \lim_{\mathbf{X}\to\mathbf{X}_0} (-f)(\mathbf{X}) &= \infty. \end{aligned}$$

$$f(x,y) = 1 - x^2 - 2y^2$$

is continuous on $\mathbb{R}^2$.

Solution: See lecture.

Example: Consider the function

$$h(x,y) = \begin{cases} \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2 < 1, \\ 1, & x^2+2y^2 = 1, \end{cases}$$

then it follows from the example we have discussed that h is continuous on the ellipse

$$x^2 + 2y^2 = 1.$$

2.13 Continuity

If $\mathbf{X}_0$ is in D_f and is a limit point of D_f, then we say that f is *continuous at* $\mathbf{X}_0$ if

$$\lim_{\mathbf{X} \to \mathbf{X}_0} f(\mathbf{X}) = f(\mathbf{X}_0).$$

Example: If

$$f(s) = \sqrt{s}$$

and

$$g(x, y) = 1 - x^2 - 2y^2,$$

then $D_f = [0, \infty]$, $D_g = \mathbb{R}^2$, and

$$T = \{(x, y) : x^2 + 2y^2 \leq 1\}.$$

We have proved that g is continuous on $\mathbb{R}^2$.

We can obtain the same conclusion by observing that the functions $p_1(x, y) = x$ and $p_2(x, y) = y$ are continuous on $\mathbb{R}^2$.

Theorem: If f is continuous on a compact set S in $\mathbb{R}^n$, then f is uniformly continuous on S.

Proof: See lecture.

Theorem: Let f be continuous on a region S in $\mathbb{R}^n$.
Suppose that $\mathbf{A}$ and $\mathbf{B}$ are in S and

$$f(\mathbf{A}) < u < f(\mathbf{B}).$$

Then $f(\mathbf{C}) = u$ for some $\mathbf{C}$ in S.

2.16.2 Equality of Mixed Partial Derivatives

Theorem: Suppose that f, f_x, f_y, and f_{xy} exist on a neighborhood N of (x_0, y_0), and f_{xy} is continuous at (x_0, y_0).

Then $f_{yx}(x_0, y_0)$ exists, and

$$f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0). \tag{2.34}$$

Since f_x and f_y are continuous on

$$S = \{(x, y) : x \neq y\},$$

then

$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = \infty$$

if $|\mathbf{X_0}| = 1$, because

$$f(\mathbf{X}) = \frac{1}{|\mathbf{X} - \mathbf{X_0}|},$$

so

$$f(\mathbf{X}) > M \quad \text{if} \quad 0 < |\mathbf{X} - \mathbf{X_0}| < \delta = \frac{1}{M}.$$

Example: If

$$\frac{\partial f(x)}{\partial \phi} = h'(0) = |x|^2 = f(x)$$

$$= \frac{\partial (x_1^2 + x_2^2)}{\partial (x_1, x_2)} = \boxed{2x_1 + 2x_2}$$

$$= (2x_1)(\phi_1) + (2x_2)\phi_2$$

$$= 2x_1\left(\frac{1}{\sqrt{2}}\right) + \frac{2x_2}{\sqrt{2}}$$

$$= \sqrt{2}x_1 + \sqrt{2}x_2$$

$$= \sqrt{2}(x_1 + x_2)$$

We have proved that g is continuous on T. Since f is continuous on D_f, the composite function $h = f \circ g$ defined by

$$h(x,y) = \begin{cases} \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2 < 1, \\ 1, & x^2+2y^2 = 1, \end{cases}$$

is continuous on

$$D_g = T = \{(x,y) : x^2 + 2y^2 \leq 1\}.$$

Example: If

$$f(x,y,z) = \cos\left(\frac{1}{x^2+2y^2+z^2}\right),$$

then

$$\lim_{|\mathbf{X}|\to\infty} f(\mathbf{X}) = 1.$$

Example: If

$$g(x,y) = \sqrt{1 - x^2 - 2y^2}$$

and

$$f(s) = \begin{cases} \frac{\sin s}{s}, & s \neq 0, \\ 1, & s = 0, \end{cases}$$

then $D_f = (-\infty, \infty)$ and

$$D_g = T = \{(x,y) : x^2 + 2y^2 \leq 1\}.$$

We have proved that g is continuous on T. Since f is cont